

MedVAE: adaptation, fine-tuning and analysis for medical imaging and coronary segmentation

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Context. Deep learning in medical imaging is limited by the high resolution required for diagnostic detail. VAEs compress the image into a compact latent.

- **MedVAE:** generic medical compressor ($> 1\text{M}$ images), $\times 16$ ($512^2 \rightarrow 128^2 \times 1$), trained in two stages (reconstruction then semantic enrichment).

Domain. Coronary angiography (ARCADE, 1 200 images): thin, tubular, low-contrast structures, noise and artifacts.

Central question

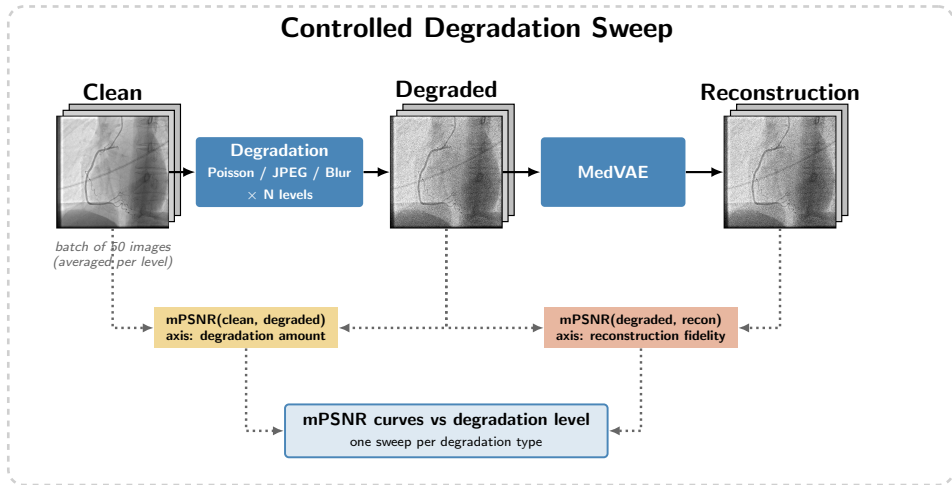
Does a generic medical compressor remain **reliable and useful** when facing a specific and demanding clinical domain?

Four complementary axes:

- 1 Robustness to degradations
- 2 Image-quality inductive bias via FiLM
- 3 Semantic structuring of the latent (JEPA)
- 4 Latent representation for segmentation

Robustness of MedVAE to input degradations

Method: evaluation protocol



Why not a no-reference metric?

Discarded approach: score the input quality with ARNIQA (no-reference), then correlate it with the reconstruction quality.

ARNIQA — clean dataset (n=1000)	
min / max	0.2695 / 0.6512
mean $\pm$ std	0.4570 $\pm$ 0.0482
quantiles 5/50/95	0.3772 / 0.4566 / 0.5391

Significant but weak correlation: the intrinsic quality variance between clean images is too small to serve as an exploitable axis.

Image propre

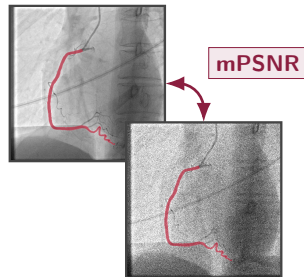
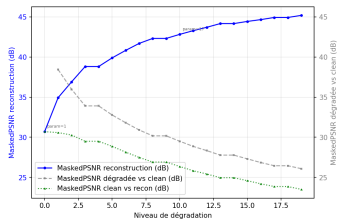


Image dégradée
(bruit de Poisson)

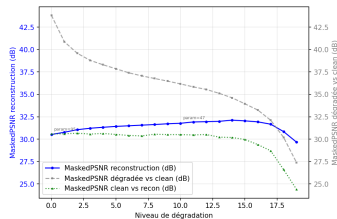
mPSNR illustration (vessel mask)

Result: three distinct regimes

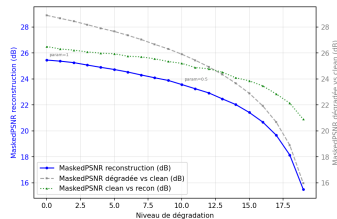
Degradation	r (Pearson)	ρ (Spearman)	
Blur	-0.998	-0.998	● Blur : near-perfect anti-correlation (28.9 → 39.2 dB) — spectral bias
Poisson	+0.998	+1.000	● Poisson : positive correlation, mirror image — the latent removes high-frequency noise (+1.94 dB toward the clean image, insufficient)
JPEG (before peak)	-0.990	-1.000	
JPEG (after peak)	+0.990	+1.000	
JPEG (global)	(+0.030)	(-0.439)	● JPEG : non-monotonic — smoothing that helps up to the peak, then collapse (block artifacts)



(a) Flou gaussien



(b) Compression JPEG

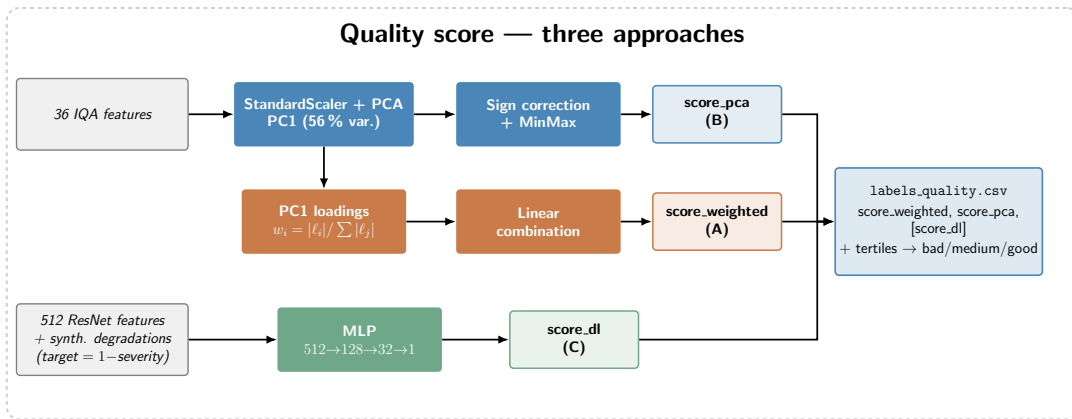


(c) Bruit de Poisson

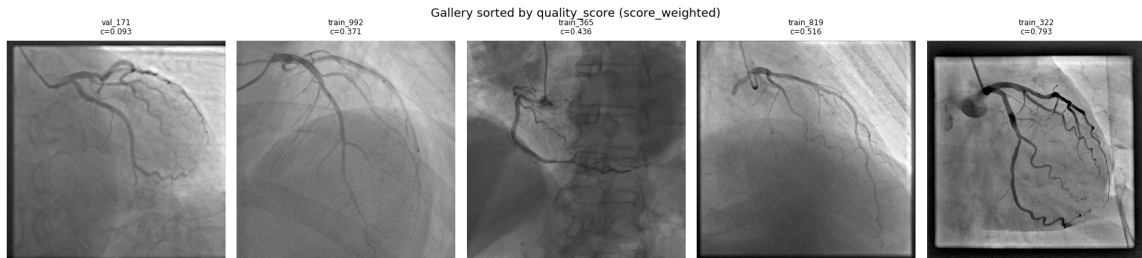
Image-quality inductive bias (FiLM)

- MedVAE applies the **same transform** to a clean or a heavily degraded coronary angiography: no notion of *how degraded* the input is.
- **Hypothesis:** conditioning MedVAE on a quality score $c \in [0, 1]$, injected via FiLM, should let it adapt to the degradation level actually present.
- Three independent ways to compute c (A: weighted, B: PCA, C: learned) $\rightarrow$ run the *same* pipeline three times.

Building the quality score c

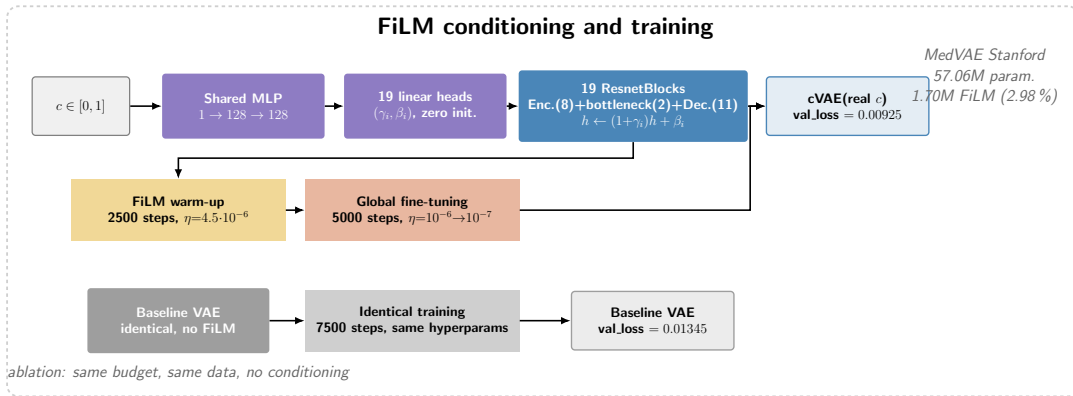


Does the score make clinical sense?

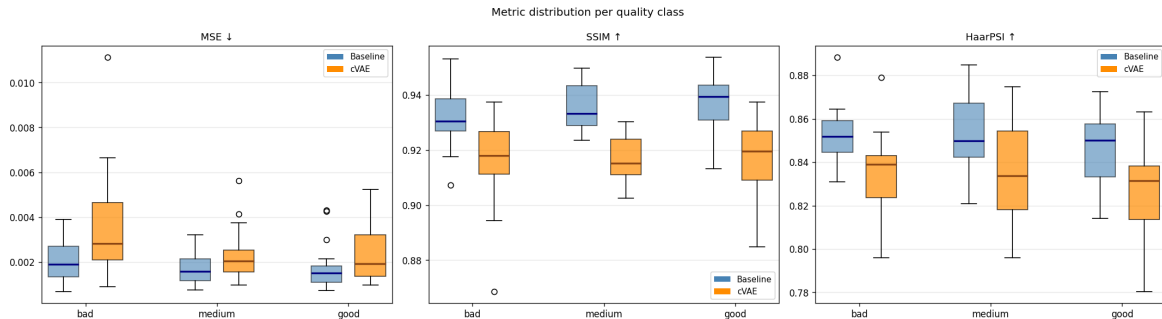


ARCADE images sorted by increasing score c (approach A).

FiLM conditioning & training



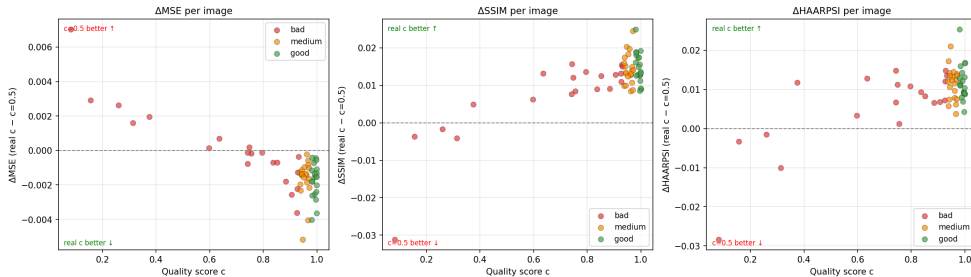
Result: conditioning hurts reconstruction



● Baseline > cVAE on **all 3 approaches** × **all 4 metrics** (MSE, SSIM, PSNR, HaarPSI), even masked

Ablation: does the network actually use c ?

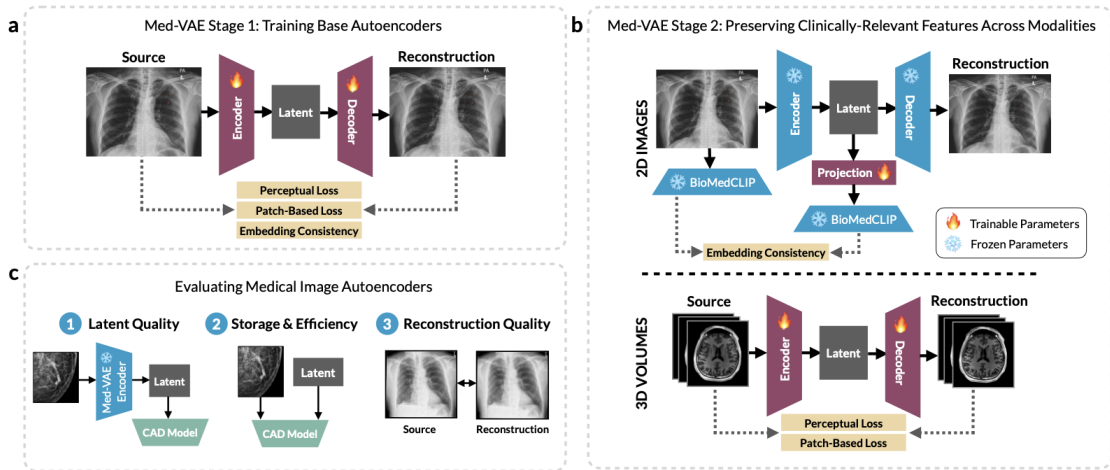
Per-image difference: cVAE(real c) vs cVAE($c=0.5$) — Approach C



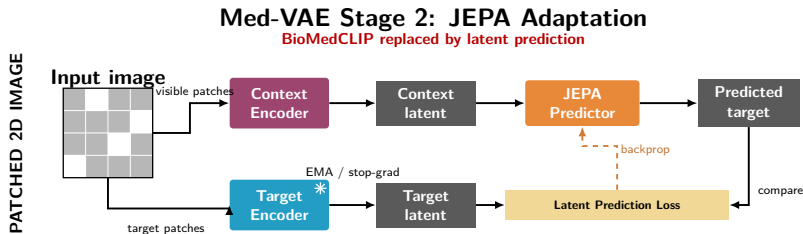
Real c vs. constant $c=0.5$ (approach C): gap grows with image quality: the network's response genuinely depends on c .

Semantic structuring of the latent space (JEPA)

Original stage 2 — BioMedCLIP distillation



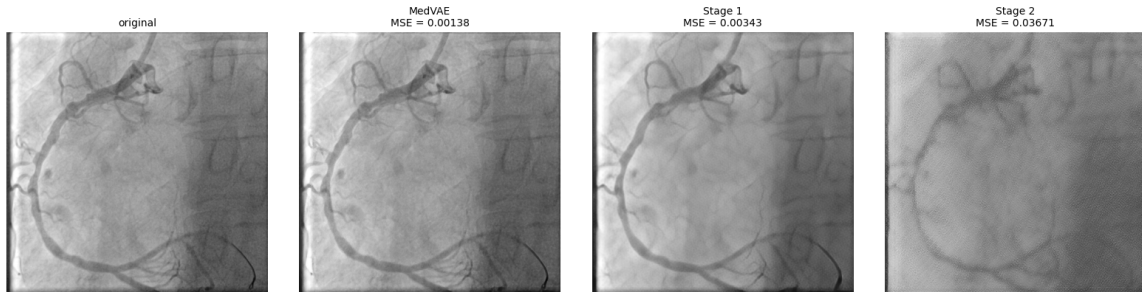
Modified stage 2 — JEPA adaptation



$$\mathcal{L}_{\text{stage2}} = \mathcal{L}_{\text{pred}} + \lambda_{\text{SIGReg}} \mathcal{L}_{\text{SIGReg}}$$

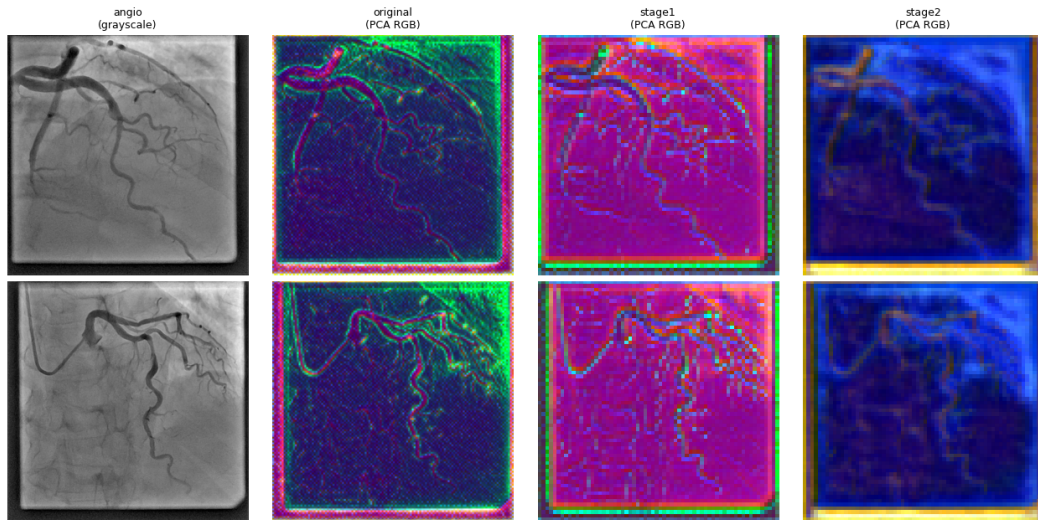
$$\mathcal{L}_{\text{pred}} = \frac{1}{|\mathcal{M}|} \sum_{p \in \mathcal{M}} \text{SmoothL}_1 \left(\hat{z}_t^{(p)}, \tilde{z}_t^{(p)} \right), \quad \mathcal{L}_{\text{SIGReg}} = \mathbb{E}_{\mathbf{v}} \left[T_{\text{EP}} \left(\{z_c^{\top} \mathbf{v}\} \right) \right]$$

Results — reconstruction



- **Frozen** decoder: any shift of the encoder degrades the PSNR (16.85 dB for stage 2).
- JEPa stage 2 **blurs**: predicting a masked patch amounts to *averaging* the plausible regions $\Rightarrow$ loss of high frequencies (thin vessels).
- Consistent with the latent objective: we optimize semantics, not pixel fidelity.

Results — RGB PCA of the intermediate features



Integration for multi-class segmentation

Task. Fine segmentation of coronary arteries on ARCADE:

- 26 classes (the background and 25 segments)
- thin, tubular, low-contrast structures
- extreme imbalance ($\approx 95\%$ background)

Tool. MedVAE, a pre-trained medical variational autoencoder: $\times 16$ compression ($512^2 \rightarrow 128^2 \times 1$), but **lossy**.

Two questions

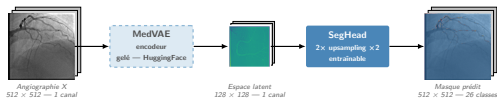
- 1 Does the compression preserve enough anatomical information for segmentation?
- 2 Where to integrate it: **latent** space or **pixel** space?

Method: five conditions

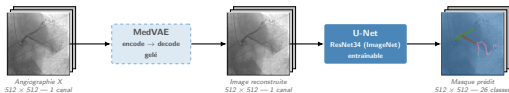
- **A** — full-resolution U-Net (reference)
- **A*** — U-Net from A on reconstructed images, **without retraining** (control)
- **B** — lightweight head on the frozen latent
- **C** — like B, encoder fine-tuned on ARCADE
- **D** — U-Net retrained on reconstructed images

Latent space: B, C ; pixel space: A*, D.

Condition B — MedVAE pré-entraîné (gelé) + tête de segmentation



Condition D — U-Net entraîné sur images reconstruites par MedVAE



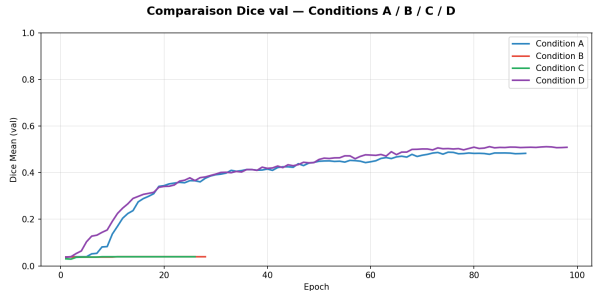
Result 1: segmentation from the latent fails

- B and C: Dice ≈ 0.04 (vs 0.433 for A)
- only the background is predicted, IoU $\rightarrow 0$

Why?

- 95 % background imbalance $\Rightarrow$ trivial minimum is a strong attractor
- single-channel latent, **no multi-scale features** (nor skip connections)

Fine-tuning the compressor (C) changes nothing.



Result 2: pixel space preserves, adaptation helps

Cond.	Dice	IoU
A	0.433	0.499
A*	0.434	0.497
B	0.038	0.005
C	0.039	0.000
D	0.451	0.495

- **Compression is neutral:** $A^* \approx A$ (0.434 vs 0.433)
- **Key result:** $D - A^* = +0.017$
pure gain from adaptation (same input, only retraining differs)
- gain concentrated on the rare, thin segments (C11, C18, C19)



Qualitative predictions

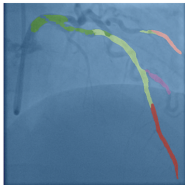
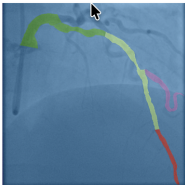
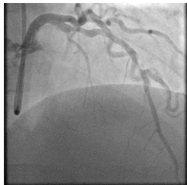
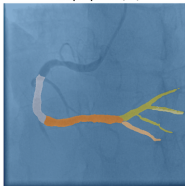
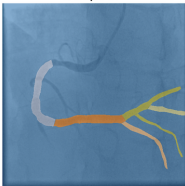
Pixel space: structures well segmented.

Condition A* — U-Net A sur images MedVAE reconstituées

Image originale

Masque GT

Masque prédit (A*)



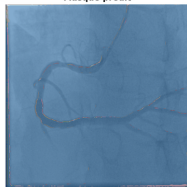
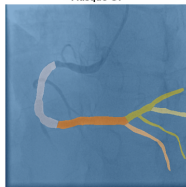
Latent space: everything is predicted as background.

Visualisation des segmentations — condition_b_medvae

Image originale

Masque GT

Masque prédit

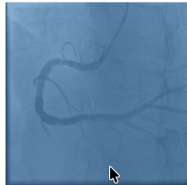
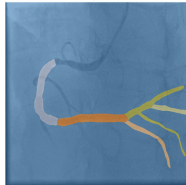


Visualisation des segmentations — condition_c_medvae_finetuned

Image originale

Masque GT

Masque prédit



Across four complementary axes, we probed the behavior of MedVAE, a generic medical compressor, on the demanding domain of coronary angiography:

- **Robustness:** intelligent low-pass filter and non-linear denoiser; collapses on *structured* high frequencies (JPEG artifacts) — the nature of the degradation matters more than its intensity.
- **FiLM:** conditioning on a quality score slightly degrades the reconstruction (architectural trade-off), even though the network genuinely exploits the signal.
- **JEPA:** markedly more compact latent (90% of variance on 25 dimensions), discriminative gain, but at a reconstruction cost (frozen decoder).
- **Segmentation:** $\times 16$ compression preserves the anatomy **in pixel space** (targeted gain on thin vessels); the single-channel latent remains unsuited to direct dense prediction.

Takeaway. Faithful compression reusable in pixel space, but the latent is not yet *semantically organized* nor *acquisition-aware* enough to be segmented or conditioned directly.

Toward a *foundation model* for radiation-based medical imaging

Unify the explored axes into a single training objective.

- **Hybrid JEPA + reconstruction objective:** predict the masked latent patches while keeping the encoder anchored to a decodable space.
- **Physically grounded degradations:** quantum noise, contrast/brightness, multiplicative fields, anisotropic motion blur (cardiac/respiratory).
- **Conditioning (CVAE)** on DICOM metadata (modality, dose, SNR, quality) → disentangled content / acquisition-quality representations.
- **Downstream validation:** classification, detection, multi-modality segmentation (ARCADE as a demanding test bed).